

CareEye: AI-Based Facial Pain Detection System

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Abstract

CareEye is an AI-based healthcare assistance system designed to detect possible pain expressions from patients' facial images in real time. The system aims to support healthcare professionals and caregivers, especially when patients are unable to clearly communicate their pain. CareEye combines computer vision and deep learning techniques with an embedded Raspberry Pi platform. A camera captures live facial images, while OpenCV is used to detect the face and preprocess the image. A Convolutional Neural Network (CNN) then classifies the facial expression into two categories: Pain and No Pain. The system performs the analysis locally on a Raspberry Pi 4 Model B, making it a practical, portable, low-cost, and privacy-aware solution for healthcare monitoring.

1. Introduction

Pain assessment is an essential part of patient care. However, some patients, such as elderly people, critically ill patients, or non-verbal individuals, may have difficulty expressing or communicating their pain. Facial expressions can provide useful information about a patient's possible discomfort.

The CareEye project was developed to provide an automatic system that can analyze facial expressions and identify possible signs of pain. The system uses artificial intelligence to process facial images and provide a prediction that can assist healthcare professionals in monitoring patients. CareEye is intended to be an assistive system and does not replace professional medical diagnosis or clinical assessment.

2. Project Objectives

The main objective of CareEye is to develop a real-time AI-based system capable of detecting possible pain expressions through facial image analysis.

- Detect the patient's face automatically from a live camera stream.
- Classify facial expressions into Pain and No Pain.
- Provide the prediction and confidence score in real time.
- Assist healthcare professionals in monitoring patients.
- Support patients who cannot clearly communicate their pain.
- Implement AI processing locally using a Raspberry Pi.
- Provide a low-cost and portable healthcare monitoring solution.
- Improve privacy by avoiding the need to send patient images to cloud servers.

3. Materials and Tools

The CareEye system uses both hardware and software components.

Hardware:

- Raspberry Pi 4 Model B: The main computing and processing unit.
- Raspberry Pi Camera Module: Used to capture live facial images and video.
- LCD Display: Used to show the classification result and confidence score.

Software:

- Raspberry Pi OS 64-bit: The operating system used on the Raspberry Pi.
- Python: Used to develop and integrate the system.
- OpenCV: Used for image processing, face detection, and real-time video capture.
- Haar Cascade Classifier: Used to detect the facial region.

- TensorFlow and Keras: Used to build and train the CNN model.
- NumPy: Used for storing and processing image data.

4. Dataset and Preprocessing

The project uses a facial pain dataset obtained from Kaggle. The selected Color Oval Frames subset was used to focus on the facial region and reduce the effect of unnecessary background information. The dataset was organized into two main classes: Pain and No Pain.

Before training the model, the images were processed using OpenCV. Each image was resized to 64×64 pixels to provide a consistent input size. The images were then converted from RGB to grayscale to simplify the input data and reduce computational requirements. After that, pixel values were normalized to a range between 0 and 1. The processed images were stored as NumPy arrays and prepared for CNN model training.

5. CNN Model

A Convolutional Neural Network (CNN) was developed using TensorFlow and Keras to classify facial images into Pain and No Pain categories.

The CNN consists of convolutional layers that extract important visual features from facial images. Max-pooling layers are used to reduce the size of the extracted feature maps and improve computational efficiency. The extracted features are then passed through dense layers for classification. The final output layer uses a sigmoid activation function because the project performs binary classification.

The model was trained using the Adam optimizer and Binary Cross-Entropy loss function. After the training process, the trained model was saved as `pain_detection_model.h5` and transferred to the Raspberry Pi for real-time use.

6. System Operation

The CareEye system operates through a simple sequence of steps:

- The Raspberry Pi Camera Module continuously captures live video frames. OpenCV receives and processes these frames. The Haar Cascade Classifier detects the patient's face within each frame.
- After detecting the face, the facial region is extracted and processed in the same way as the training images. The image is resized to 64×64 pixels, converted to grayscale, and normalized between 0 and 1.
- The preprocessed facial image is then passed to the trained CNN model. The model analyzes the facial features and produces one of two predictions: Pain or No Pain. The system also provides a confidence score for prediction. The result can be displayed on an LCD screen to provide immediate information to the healthcare professional or caregiver.

7. System Architecture

The overall workflow of CareEye can be summarized as follows:

Camera → Live Video Frame → Face Detection → Face Extraction → Image Preprocessing → CNN Model → Pain/No Pain Prediction → Confidence Score → LCD Display

The system combines computer vision and deep learning in an embedded environment. OpenCV and Haar Cascade are responsible for detecting and preparing the facial region, while the CNN model performs the pain classification.

8. Edge Computing and Privacy

One of the main features of CareEye is that the AI model runs locally on the Raspberry Pi 4 Model B. This approach is known as edge computing, where data is processed near the source instead of being continuously sent to a cloud server.

Local processing provides several benefits. It can reduce the delay between capturing an image and producing a prediction. It also helps protect patient privacy because facial images can be processed locally without requiring continuous transmission to external servers. Furthermore, the system can operate without depending on a continuous internet connection.

Using Raspberry Pi also makes the system relatively affordable and portable, which can make it suitable for different healthcare environments.

9. Applications

CareEye can be used as an assistive monitoring system in several situations, including:

- Hospitals and patient wards.
- Intensive Care Units (ICUs).
- Elderly care facilities.
- Nursing homes.
- Home healthcare environments.
- Monitoring non-verbal patients.
- Patients with communication difficulties.
- Smart healthcare systems.

The system can provide an additional indication of possible patient discomfort and help caregivers pay attention to patients who may require further assessment.

10. Results and Discussion

The CareEye project demonstrates the integration of artificial intelligence, computer vision, and embedded systems for real-time facial pain detection. The trained CNN model is used to classify facial expressions into Pain and No Pain categories, while OpenCV and Haar Cascade provide real-time face detection and image preprocessing.

The deployment of the model on the Raspberry Pi demonstrates the possibility of performing AI-based facial analysis on a small embedded device rather than relying on a high-performance computer or cloud platform. The system provides a prediction and confidence score that can assist healthcare professionals during patient monitoring.

However, facial expressions can vary between individuals, and factors such as lighting, camera angle, image quality, and facial orientation may affect the prediction. Therefore, the system should be considered an assistive tool and not a replacement for professional medical evaluation.

11. Future Work

Future improvements to CareEye may include training the model using a larger and more diverse dataset to improve its generalization. The system could also be developed to recognize different levels of pain instead of only Pain and No Pain. Another possible improvement is combining facial expressions with other physiological information, such as heart rate or other vital signs, to provide a more comprehensive assessment.

The system could also be improved by using more advanced deep learning architectures and developing a user-friendly interface that allows healthcare professionals to monitor multiple patients and record prediction results.

12. Conclusion

CareEye is an AI-based healthcare assistance system designed to detect possible pain expressions through real-time facial image analysis. The project combines a Raspberry Pi 4 Model B, Raspberry Pi Camera Module, OpenCV, Haar Cascade face detection, and a CNN model developed using TensorFlow and Keras.

The system processes facial images locally and classifies them into Pain or No Pain categories, providing a confidence score that can assist healthcare professionals in patient monitoring. CareEye is particularly useful for patients who may have difficulty communicating their pain, such as elderly, non-verbal, or critically ill patients.

Overall, the project demonstrates how artificial intelligence, computer vision, and edge computing can be combined to create a practical, portable, and privacy-aware solution for healthcare assistance and patient monitoring. In the following page, some figures are shown from our project:

